ST5202 — Applied Regression Analysis
Lecture Notes 1. Simple Linear Regression

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Outline

§1. Introduction to the fundamental

§2. The Estimation of LRM

§3. Theoretical Properties of LSE

§4. Statistical Inference for simple LRM

§1.1. A little bit philosophy

- Everything in the universe is changing.
- The change of one thing is caused by the change of something else. The whole universe is an interconnected system of cause-and-effect chains.
- Things which change concurrently but do not have a cause-and-effect relation might be related to each other due to the same causes.
- The change of a particular thing, say y, can be explained either by the change of its cause or by the change of related concurrent things, say $\boldsymbol{x}$. This relation can be expressed as

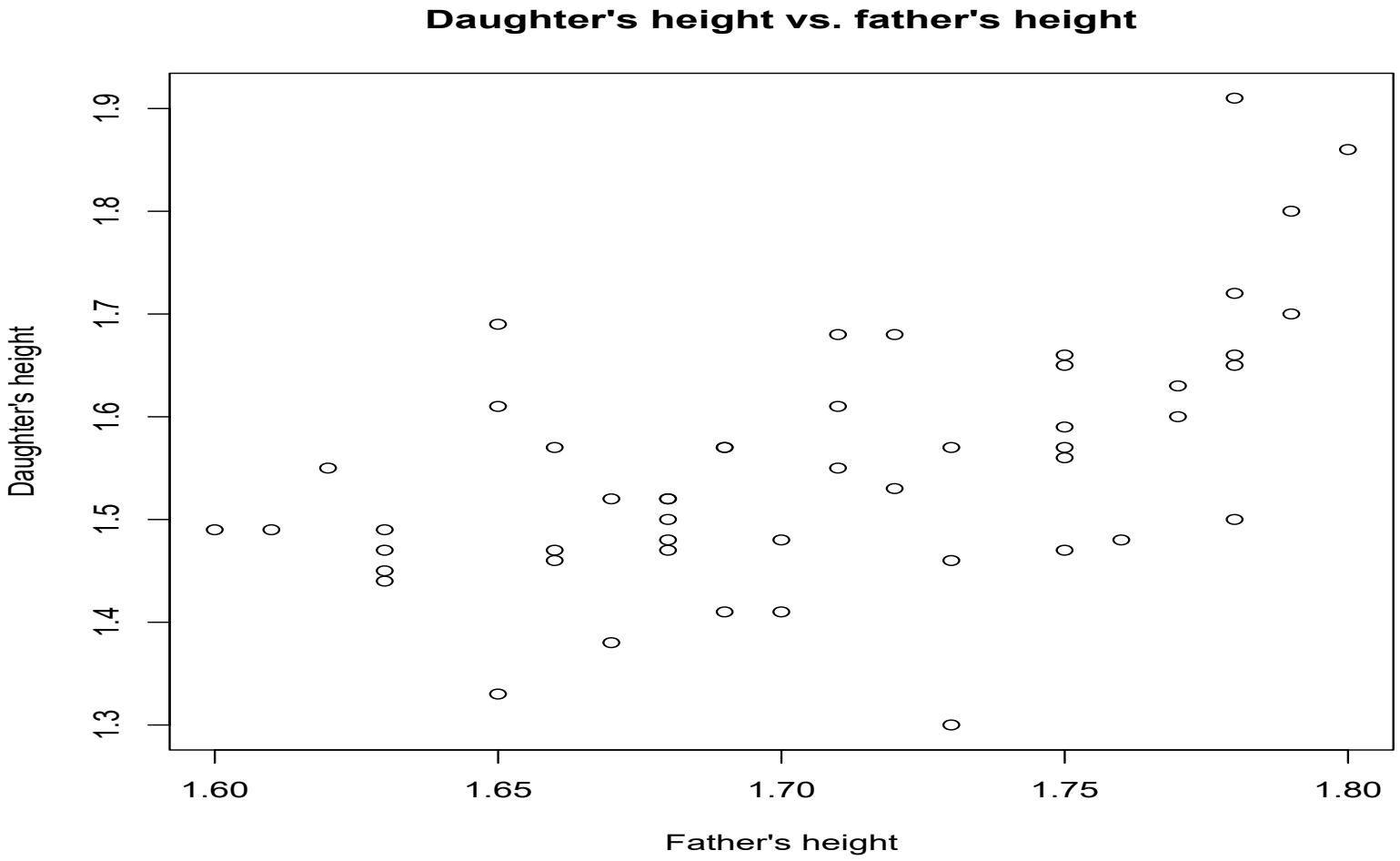
$$y = f(\boldsymbol{x}) + \epsilon.$$

- Regression analysis provides a tool to unveil the above relationship.

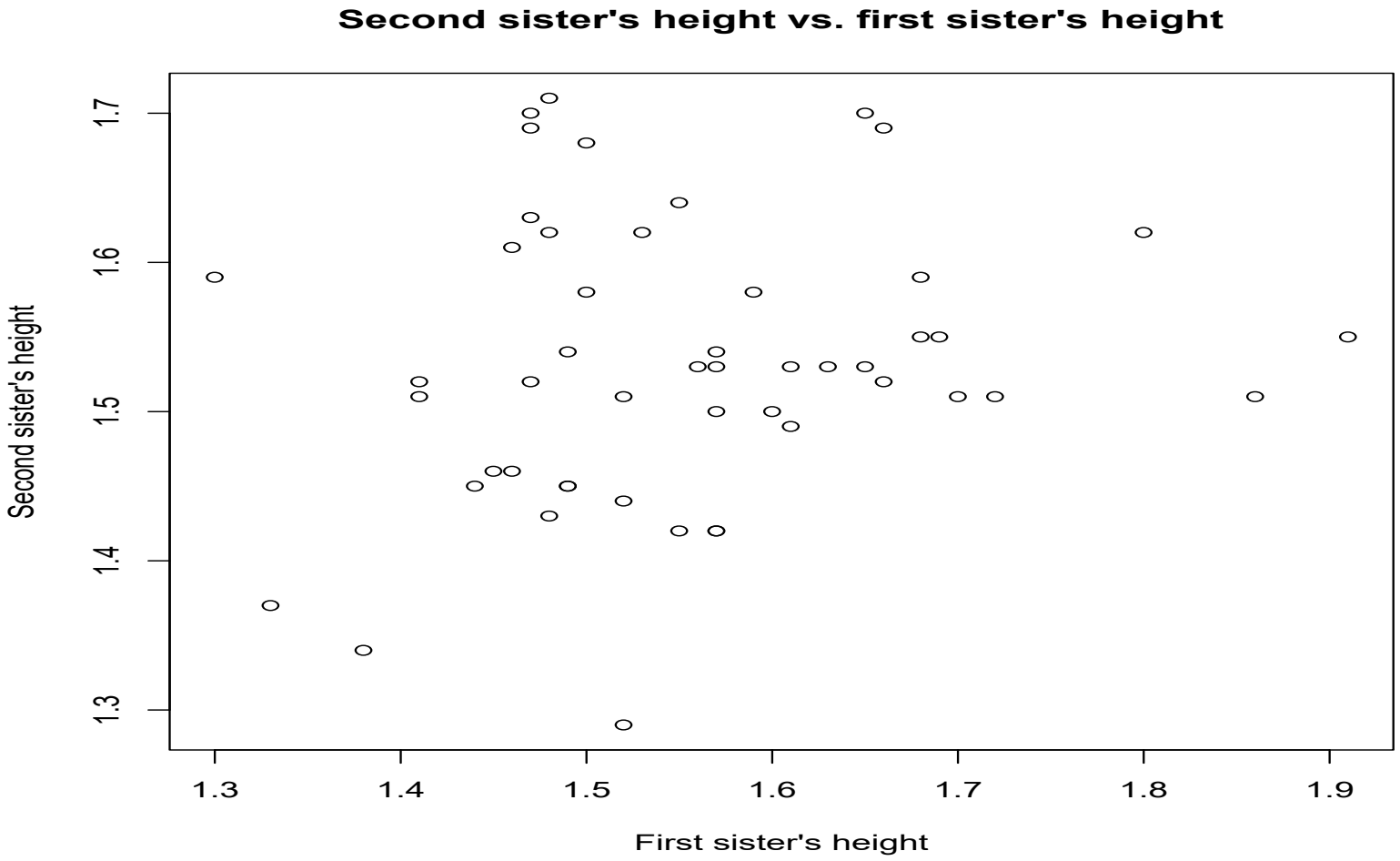
§1.2. Example of causal and non-causal relation

- On the following two slides, plotted are (1) daughter's hight against father's height and (2) the second sister's hight against the first sister's highter.
- In both plots, a linear trend show up, indicating there are a relationship between the two variables.
- In the case of daughter's and father's height, the relationship is cause-and-effect; in the case of first sister's and second sister's height, the relationship is non-causal, the two variables are concurrently changing as father's height changes.

§1.3. Daughter's height versus father's height



§1.4. Second sister's height versus first sister's height



§1.5. Correlation

- In the previous example, daughter's height is not completely determined by the father's height, there are additional factors that affect daughter's height as well. How strongly do the father's height affect the daughter's height in a population can be measured by the Pearson's correlation coefficient (or simply Pearson's correlation).
- Let X and Y be two random variables, the theoretical Pearson's correlation is defined as

$$\rho_{xy} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}.$$

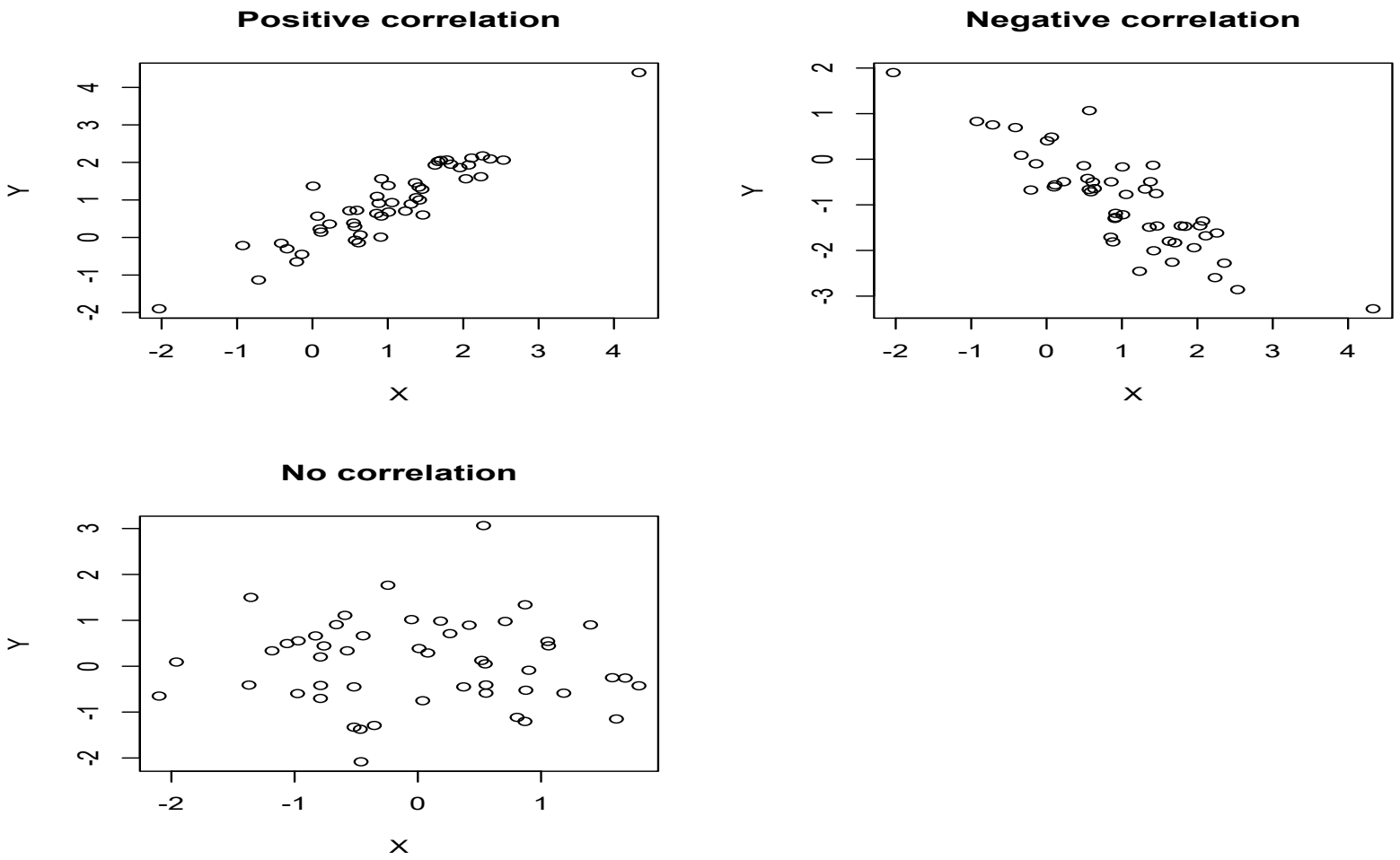
- If (X, Y) are observed on n subjects with the observations $(x_i, y_i), i = 1, \ldots, n$, the sample Pearson's correlation is given by

$$\hat{\rho}_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}.$$

§1.6. Interpretation of Correlation

- The value of Pearson's correlation coefficient is between -1 and 1, i.e., $-1 \leq \rho_{xy} \leq 1$.
- If $\rho_{xy} > 0$, then one variable tends to have a larger value while the other one has a larger value.
- If $\rho_{xy} < 0$, then one variable tends to have a smaller value while the other one has a larger value.
- If $\rho_{xy} = 0$, Y and X have no linear relationship.
- Non-zero correlation does not imply that the variables have a causal relationship, e.g., in the previous plotted examples, the heights of two sisters have a correlation 0.159, but there is no causal relation between the heights of the two sisters.
- Zero correlation does not imply the two variables have no relationship.

§1.7. Illustration of correlations



§1.8. The regression framework

- The relationship between two variables like farther's height and daughter's height can be explicitly described by the simple linear regression model (LRM):

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

where

(i) Y is the variable which we are seeking the explanation of its variation is called the response variable.
(ii) X which is the variable affecting Y is called the covariate (or predictor).
(iii) β_0, β_1 are called regression coefficients (or parameters).
(iv) ϵ is a random error which accounts for the effect by other factors affecting Y.

- The function $Y(X) = \beta_0 + \beta_1 X$ is called the regression function.

§1.9. The goal of regression analysis

A regression analysis is to establish a relationship between the response variable and the covariate for the following purposes:

- Interpreting the variation of the response variable, i.e., how much of its variation is explained by the covariate;
- Predicting the response variable from the covariate;
- Controlling the value of the response by manipulating the covariate, if the relationship is causal.

§1.10. The sample simple LRM

- The regression relationship is a population characteristic.
- It is investigated through a random sample. Suppose we have taken a random sample of size n: $(Y_1, X_1), \ldots, (Y_n, X_n)$. The sample simple LRM is given by

$$\begin{aligned} Y_1 &= \beta_0 + \beta_1 X_1 + \varepsilon_1, \\ Y_2 &= \beta_0 + \beta_1 X_2 + \varepsilon_2, \\ &\vdots \\ Y_n &= \beta_0 + \beta_1 X_n + \varepsilon_n. \end{aligned}$$

or, briefly,

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, 2, ..., n.$$

§1.11. Assumptions on the simple LRM

- General assumptions:
  - (i) The X_i and ϵ_i are independent.
  - (ii) The ϵ_i's have mean zero.
  - (iii) The ϵ_i's are pairwise un-correlated, i.e., $\text{Cov}(\epsilon_i, \epsilon_j) = 0$.
  - (iv) The ϵ_i's have common variance σ^2 (Homogeneity).
  - (v) The ϵ_i's have a Normal distribution (Normality).
- When all the conditions above are satisfied, the LRM is called a Normal LRM. The assumptions for a Normal LRM can be simply stated as, in addition to (i), ϵ_i's are are i.i.d. $N(0, \sigma^2)$ variables.
- In practice, the Normality is not necessarily required. The most important assumption is the Homogeneity.

§2.1. The meaning of the regression coefficients

- In the regression model $Y = \beta_0 + \beta_1 X + \epsilon$, when X and ϵ are independent, we have
$$\mathrm{Cov}(X, Y) = \beta_1 \mathrm{Var}(X), \quad EY = \beta_0 + \beta_1 EX.$$
- From the above relationship, we have
$$\beta_1 = \rho_{xy} \frac{\sigma_y}{\sigma_x}, \ \beta_0 = \mu_y - \beta_1 \mu_x,$$
where ρ_{xy} is the Pearson's correlation coefficient between Y and X, σ_x and σ_y are the standard deviations of X and Y respectively, μ_x and μ_y are the means of X and Y respectively. Essentially, β_1 measures the correlation between X and Y.
- It can be verified that (β_0, β_1) is the solution to
$$\min_{b_0, b_1} E_F (Y - b_0 - b_1 X)^2,$$
where F is the joint distribution function of (Y, X). The regression function $\beta_0 + \beta_1 X$ is referred to as the best linear approximation of X to Y.

§2.2. Least Square Estimation

- By the empirical principle, the parameters (β_0, β_1) are estimated by their sample counterparts, i.e., the minimizer of $E_{F_n}(Y - b_0 - b_1 X)^2$, where F_n is the empirical distribution.

$$E_{F_n}(Y - b_0 - b_1 X)^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2,$$

  where $(y_1, x_1), \ldots, (y_n, x_n)$ are observed values of $(Y_1, X_1), \ldots, (Y_n, X_n)$
- This gives rise to the least squares estimation (LSE) which estimates the parameters by minimizing the sum of squares

$$Q = \sum_{i=1}^{n}(y_i - b_0 - b_1 x_i)^2.$$

- Remark: Conceptually, (β_0, β_1) are unknown but fixed parameters, and b_0, b_1 are variables. In what follows, by an abuse of notation, b_0, b_1 will be replaced by β_0, β_1 and denote Q as

$$Q = \sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2.$$

§2.3. The LSE of β_0 and β_1

- By calculus, we have

$$\frac{\partial Q}{\partial \beta_0} = -2\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i), \quad \frac{\partial Q}{\partial \beta_1} = -2\sum_{i=1}^{n} x_i(y_i - \beta_0 - \beta_1 x_i).$$

Solving the normal equations:

$$-2\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i) = 0, \quad -2\sum_{i=1}^{n} x_i(y_i - \beta_0 - \beta_1 x_i) = 0,$$

we have the LSE of β_0 and β_1:

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^{n}(x_i - \bar{X})(y_i - \bar{Y})}{\sum_{i=1}^{n}(x_i - \bar{X})^2} = \hat{\rho}_{xy}\frac{\hat{\sigma}_y}{\hat{\sigma}_x}, \\ \hat{\beta}_0 &= \bar{Y} - \hat{\beta}_1\bar{X} = \hat{\mu}_y - \hat{\beta}_1\hat{\mu}_x, \end{aligned}$$

where $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$, $\bar{Y} = \frac{1}{n}\sum_{i=1}^{n} y_i$.

- Remark: The $\hat{\beta}_0$ and $\hat{\beta}_1$ are in fact the sample counterparts of β_0 and β_1.

§2.4. The estimate of σ^2

- The σ^2 is estimated by $\hat{\sigma}^2 = SSE/(n-2)$, where
$$SSE = \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^{n}\hat{\epsilon}_i^2, \quad \hat{\epsilon}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i.$$
- The rationale:
  - Since $\sigma^2 = E\epsilon^2$ (because $E\epsilon = 0$), if ϵ_i's were observable, σ^2 could be estimated by its sample counterpart $\frac{1}{n}\sum_{i=1}^{n}\epsilon_i^2$, which provides an unbiased estimate.
  - But ϵ_i's are not observable, ϵ_i can only be approximated by $\hat{\epsilon}_i$. Replacing ϵ_i by $\hat{\epsilon}_i$, we have the estimate $\tilde{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\hat{\epsilon}_i^2$.
  - The $\tilde{\sigma}^2$ is not unbiased since the $\hat{\epsilon}_i$'s are not independent, they are subject to two constraints: $\sum_{i=1}^{n}\hat{\epsilon}_i = 0$, $\sum_{i=1}^{n}x_i\hat{\epsilon}_i = 0$. This means that the n $\hat{\epsilon}_i$'s only accounts for the effect of $n-2$ independent error terms. To adjust the bias, the divisor n in $\tilde{\sigma}^2$ is replaced by $n-2$.
- The estimate $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ is called the residual standard error.

§2.5. The LSE as Maximum Likelihood estimation

- Under the Normality assumption, the log-likelihood is

$$-\frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1 x_i)^2 = -\frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}Q.$$

- Maximizing the log-likelihood to obtain the MLE of the regression coefficients is equivalent to minimizing Q, the sum of squares. Thus the MLE of β_0, β_1 are the same as those of the LSE.
- The MLE of σ^2, $\hat{\sigma}^2_{\mathrm{MLE}}$, and $\hat{\sigma}^2$ have the relation:

$$\hat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \frac{n-2}{n}\hat{\sigma}^2.$$

§2.6. Terminology used in regression analysis

- The fitted (or estimated) regression function:

$$\hat{Y}(X) = \hat{\beta}_0 + \hat{\beta}_1 X.$$

  For convenience, $\hat{Y}(X)$ will be denoted simply as $\hat{Y}$ from now onward.

- The fitted values,

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \qquad i = 1, ..., n.$$

- The residuals,

$$e_i = y_i - \hat{y}_i, \ \ i = 1, 2, ..., n.$$

- For a new subject with $X = X^*$, the value $\hat{Y}^* = \hat{\beta}_0 + \hat{\beta}_1 X^*$ is called the predicted value at X^*.

§2.7. Decomposition of total sum of squares

- The variation of Y is caused by two sources: the effect of X and the effect of random errors.
- The total variation of Y is estimated by $\text{SST} = \sum_{i=1}^{n}(y_i - \bar{Y})^2$, which is called the total sum of squares. It can be decomposed as

$$\text{SST} = \sum_{i=1}^{n}(\hat{y}_i - \bar{Y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \equiv \text{SSR} + \text{SSE},$$

  where SSR called the regression sum of squares accounts for the variation explained by X, and SSE called the residual sum of squares accounts for the variation caused by random errors.
- ANOVA table:

Source	DF	SS	MS	F
Regression	1	SSR	MSR	MSR/MSE
Error	$n-2$	SSE	MSE	
Total	$n-1$	SST		

- The verification of the decomposition and the meaning of DF will be provided later.

§2.8. Coefficient of determination R^2

- From the decomposition of total sum of squares, we can define the coefficient of determination as follows:

$$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{Y})^2}{\sum_{i=1}^{n}(y_i - \bar{Y})^2} = \frac{\text{SSR}}{\text{SST}}.$$

- R^2 is the proportion of the variation in Y explained by X in the sample. It measures the strength of the correlation between Y and X.
- It can be verified that

$$R^2 = \text{corr}(Y, X)^2 = \text{corr}(Y, \hat{Y})^2, \tag{1}$$

where $\text{corr}(Y, \hat{Y})$ is the sample Pearson's correlation between Y and $\hat{Y}$ and $\text{corr}(Y, X)$ is the sample Pearson's correlation between Y and X.

§2.9. (optional) Verification

Since $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i = \bar{Y} + \hat{\beta}_1(x_i - \bar{X})$, $\bar{\hat{Y}} = \frac{1}{n}\sum_{i=1}^n \hat{y}_i = \bar{Y}$ and $\hat{y}_i - \bar{\hat{Y}} = \hat{y}_i - \bar{Y} = \hat{\beta}_1(x_i - \bar{X})$. Hence,

$$\sum_{i=1}^n (\hat{y}_i - \bar{Y})^2 = \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{X})^2 = \frac{[\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})]^2}{\sum_{i=1}^n (x_i - \bar{X})^2},$$

and

$$\begin{aligned}
\frac{\sum_{i=1}^n (\hat{y}_i - \bar{Y})^2}{\sum_{i=1}^n (y_i - \bar{Y})^2} &= \frac{[\sum_{i=1}^n (x_i - \bar{X})(y_i - \bar{Y})]^2}{\sum_{i=1}^n (x_i - \bar{X})^2 \sum_{i=1}^n (y_i - \bar{Y})^2} = \mathrm{corr}(Y, X)^2 \\
&= \frac{[(1/\hat{\beta}_1)\sum_{i=1}^n (\hat{y}_i - \bar{\hat{Y}})(y_i - \bar{Y})]^2}{(1/\hat{\beta}_1^2)\sum_{i=1}^n (\hat{y}_i - \bar{\hat{Y}})^2 \sum_{i=1}^n (y_i - \bar{Y})^2} \\
&= \mathrm{corr}(Y, \hat{Y})^2
\end{aligned}$$

§2.10. Adjusted R^2

- The R^2 is the proportion of the variation of Y in the sample explained by X. As an estimation of the population counterpart, it is not unbiased.
- A less biased estimate is given by the Adjusted R^2:

$$R_a^2 = \frac{n-1}{n-p-1} R^2 - \frac{p}{n-p-1},$$

  where p is the number of predictors and equals 1 for simple LRM.
- R^2 is strictly increasing as the number of predictors, p, increases. But R_a^2 does not necessarily increase when p increases. Historically, R_a^2 is used as a model selection criterion.

§2.11. Usage of a fitted LRM

After fitting a simple LRM, we have the estimated parameters $\hat{\beta}_0$, $\hat{\beta}_1$, the R^2 and the estimated regression function:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X.$$

- By using the R^2, we can explain how much of the variation of Y can be attributed to X.
- When the relationship is causal, we can control the value of Y by manipulating the value of X. $\hat{\beta}_1$ is the estimated amount of change in the expectation of Y when X increases by an unit amount.
- When we have a new observation $X = X^*$ but the corresponding Y^* is not yet observed, we can predict Y^* by $\hat{\beta}_0 + \hat{\beta}_1 X^*$.

§2.12. Illustration example (1)

- The data:

Obs.	1	2	3	4	5	6
X	1.0	1.5	2.1	2.9	3.2	3.9
Y	0.60	2.00	1.06	3.44	1.17	3.54

- Calculation:

$$\bar{X} = 2.4333, \quad \bar{Y} = 1.9683,$$

$$\sum_{i=1}^{6}(x_i - \bar{X})^2 = 5.9933,$$

$$\sum_{i=1}^{6}(x_i - \bar{X})(y_i - \bar{Y}) = 4.6143.$$

$$\hat{\beta}_1 = \frac{4.6143}{5.9933} = 0.7699,$$

$$\hat{\beta}_0 = 1.9683 - 2.4333 * 0.7699 = 0.0949.$$

§2.13. Illustration example (2)

$$SST = \sum_{i=1}^{6}(y_i - \bar{Y})^2 = 7.9717,$$

$$SSR = \hat{\beta}_1^2 \sum_{i=1}^{6}(x_i - \bar{X})^2 = 0.7699^2 * 5.9933 = 3.5525.$$

$$SSE = 7.9717 - 3.5525 = 4.4192, R^2 = \frac{3.5525}{7.9717} = 0.4456.$$

- The proportion of the variation in Y explained by X is 0.4456.
- If we change X to $X + 1$, Y is expected to have a estemated increase 0.7699.
- Suppose $X = 3$ is a new observed value of X, the corresponding Y value is predicted as

$$\hat{Y} = 0.0949 + 0.7699 * 3 = 2.4046.$$

§2.14. Regression analysis by using R

- `read.csv`: input data from Excel file with `.csv` format.
- `read.table`: input data from text file with `.txt` format.
- `plot(x, y)`: plot `y` against `x`.
- `fitted.object = lm(y~x)`: to call the `lm` to estimate the model and store the results in `fitted.object`.
- `summary(fitted.object)`: list the estimation results.
- `anova(fitted.object)`: obtain ANOVA table.
- `abline(fitted.object)`: draw the fitted regression line over the scatter plot of `y` againt `x`, it must be preceded by `plot(x,y)`.
- `yhat = fitted.object$fitted`: extract the fitted values and store them in `yhat`.
- `r=fitted.object$resid`: extract the residuals and store them in `r`.
- Predicting at new covariate values:

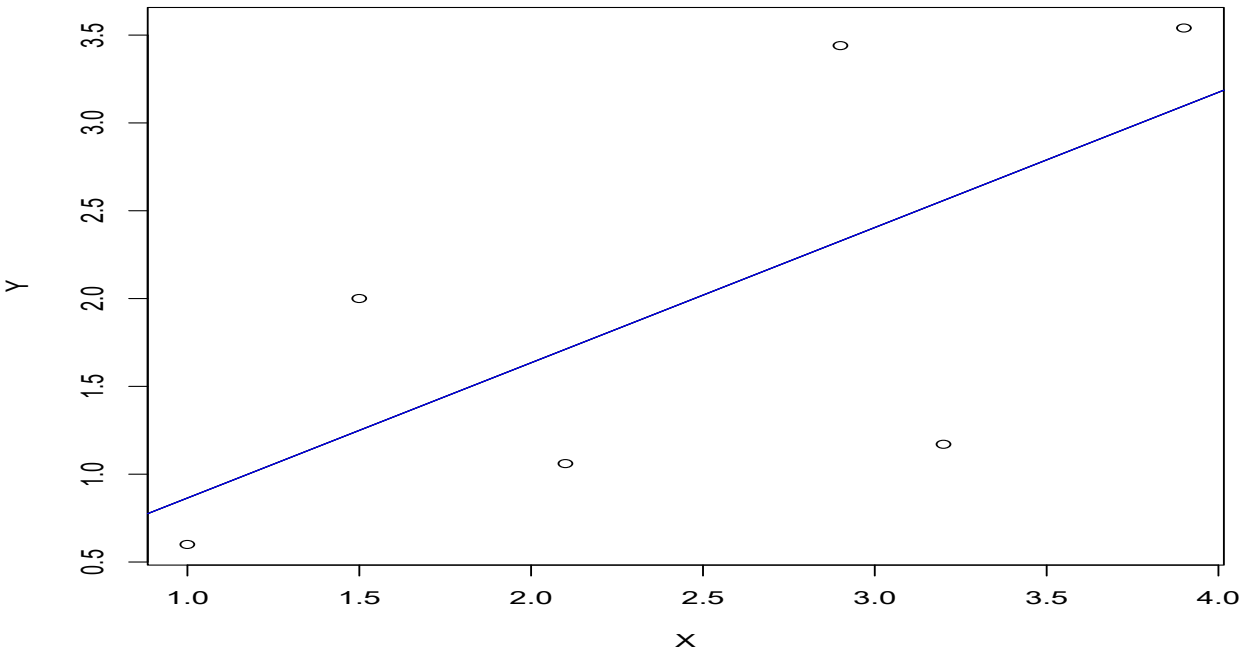
```
predict(fitted.object, newdata,
        interval = c("none", "confidence", "prediction"),
        level = 0.95)
```

§2.15. Illustration example (3)

The computation for the illustration example can be carried out by the following R code:

```
X = c(1.0, 1.5, 2.1, 2.9, 3.2, 3.9)
Y = c(0.60, 2.00, 1.06, 3.44, 1.17, 3.54)
plot(X, Y)  # plot the observations (data)
y.fit = lm(Y~X)
abline(y.fit)
summary(y.fit)
residuals = y.fit$resid
fitted = y.fit$fitted
```

§2.16. Scatter plot and fitted regression line



§2.17. Results produced by the R codes

Summary results:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.09488    1.12942   0.084    0.937
X            0.76991    0.42934   1.793    0.147
Residual standard error: 1.051 on 4 degrees of freedom
Multiple R-squared:  0.4457,
Adjusted R-squared: 0.3071
F-statistic: 3.216 on 1 and 4 DF,
p-value: 0.1474
```

§2.18. Interpretation of Summary results

- `Estimate`: (Intercept) — $\hat{\beta}_0$, X — $\hat{\beta}_1$.
- `Std. Error`: $sd(\hat{\beta}_0)$, $sd(\hat{\beta}_1)$.
- `t value`: $\hat{\beta}_0/sd(\hat{\beta}_0)$, $\hat{\beta}_1/sd(\hat{\beta}_1)$.
- `Pr(>|t|)`: p-values for two-sided tests on β_0 and β_1.
- `Residual standard error`: $\hat{\sigma}$.
- `Multiple R-squared`: R^2.
- `Adjusted R-squared`: R_a^2.
- `F-statistic`: MSR/MSE.
- `p-value`: p-value of the significant F test.

§2.19. Muscle mass (MM) example

A person's muscle mass is expected to decrease with age. To explore this relationship in women, a nutritionist randomly selected four women from each 10-year age group, beginning with age 40 and ending with age 79. The data is given in `MascleMass.txt` on Canvas. The results are as follows: X age, Y a measure of muscle mass:

i:	1	2	$\cdots$	15	16
x_i	71	64	$\cdots$	49	78
y_i	82	91	$\cdots$	105	77

(i) Write R codes for fitting a LRM to the data.

(ii) Check whether the fitted model is appropriate by the plots: (a) scatter plot of y vs. x with fitted regression line imposed, (b) plot of residuals vs. fitted values.

(iii) Obtain the fitted regression function and to see whether a person's muscle mass is expected to decrease with age.

(iv) What is the proportion of the variation in muscle mass which is explained by age?

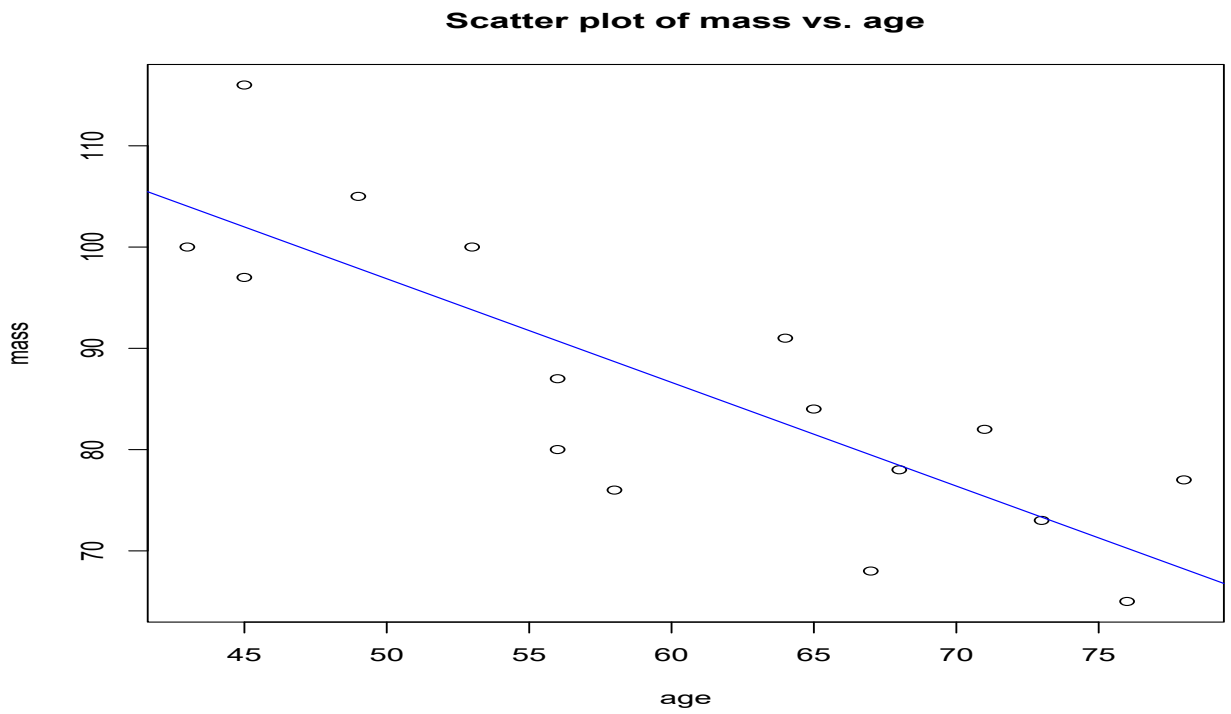
§2.20. R codes

```
# Input data
  MM.dat = read.table(file="D://Rsession/MuscleMass.txt",
                      header=TRUE)
  mass=MM.dat[,1]; age=MM.dat[,2]

# Fit the model
  MM.fit=lm(mass~age);  summary(MM.fit)

# Plots for checking model adequacy
  plot(age, mass, main="Scatter plot of mass vs. age")
  abline(MM.fit)

  r=MM.fit$resi;  fitted = MM.fit$fitted
  plot(fitted,r,main="Residual plot v.s. fitted values")
```

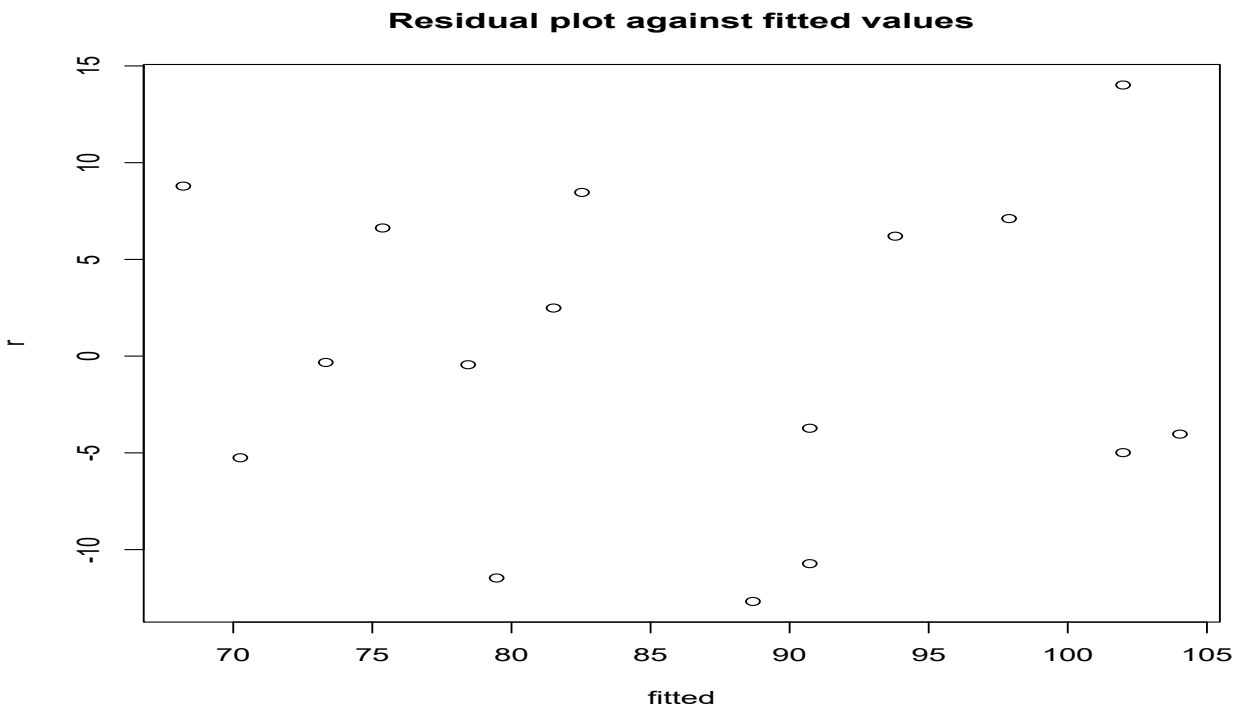
§2.21. Fitted results

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 148.0507    11.5629  12.804 4.05e-09
age          -1.0236     0.1882  -5.439 8.72e-05

Residual standard error: 8.344 on 14 degrees of freedom
Multiple R-squared: 0.6788,
Adjusted R-squared: 0.6559
F-statistic: 29.59 on 1 and 14 DF,
p-value: 8.721e-05
```

§2.22. Scatter plot with fitted regression function



§2.23. Residual plot



§2.24. Conclusion

- The data points are quite evenly scattered on both sides of the fitted regression line, which suggests that the simple LRM is appropriate. The points in the residual plot are scattered quite evenly within a horizontal band, which suggests that the variances are constant. It can be concluded that the fitted model is appropriate.
- The fitted regression function is given by

$$\hat{Y} = 148.0507 - 1.0236X.$$

  The coefficient of X is negative, which indicates that muscle mass is expected to decrease with age.
- The proportion of the total variation in muscle mass explained by age is given by $R^2 = 0.6788$.

§3.1. A general remark on regression analysis

- In a regression model, both the response variable Y and the covariate X are random variables.
- However, the regression analysis is a conditional analysis, that is, it is conditioning on the observed values of the covariate. In other words, the individual X_i's are treated as non-random variables.
- Thus, for given X, the expectation of Y is $\beta_0 + \beta_1 X$ instead of $\beta_0 + \beta_1 EX$.
- Throughout, all statement about distributions and the related quantities are conditional ones, e.g., $E\hat{\beta}_1$ and $\mathrm{Var}(\hat{\beta}_1)$ are indeed $E(\hat{\beta}_1|X)$ and $\mathrm{Var}(\hat{\beta}_1|X)$. For the sake of convenience, in the notation, the condition will be omitted.

§3.2. Unbiasedness of $\hat{\beta}_0$ and $\hat{\beta}_1$

- The LSE is unbiased, i.e., $E\hat{\beta}_1 = \beta_1$, $E\hat{\beta}_0 = \beta_0$.
  Proof:

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \frac{\sum_{i=1}^n (X_i - \bar{X})(\beta_0 + \beta_1 X_i + \epsilon_i)}{\sum_{i=1}^n (X_i - \bar{X})^2} \\
&= \frac{\sum_{i=1}^n (X_i - \bar{X})(\beta_1 X_i + \epsilon_i)}{\sum_{i=1}^n (X_i - \bar{X})^2} = \beta_1 + \frac{\sum_{i=1}^n (X_i - \bar{X})\varepsilon_i}{\sum_{i=1}^n (X_i - \bar{X})^2}.
\end{aligned}$$

  Hence $E\hat{\beta}_1 = \beta_1$.
  By summing up $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, we have

$$\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{\epsilon},$$

  where $\bar{\epsilon} = \sum_{i=1}^n \epsilon_i / n$. Hence,

$$E(\hat{\beta}_0) = E(\bar{Y} - \hat{\beta}_1 \bar{X}) = E(\beta_0 + \beta_1 \bar{X} + \bar{\epsilon} - \hat{\beta}_1 \bar{X}) = \beta_0,$$

  noting that $E\hat{\beta}_1 = \beta_1$ and $E\bar{\epsilon} = 0$.

§3.3. The variance of $\hat{\beta}_0$ and $\hat{\beta}_1$

$$
\begin{aligned}
\mathrm{Var}(\hat{\beta}_1) &= Var\left(\frac{\sum_{i=1}^n (X_i - \bar{X})Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2}\right) \\
&= \left[\sum_{i=1}^n (X_i - \bar{X})^2\right]^{-2} \sum_{i=1}^n (X_i - \bar{X})^2 \sigma^2 = \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}. \\
\mathrm{Var}(\hat{\beta}_0) &= \mathrm{Var}(\bar{Y} - \hat{\beta}_1 \bar{X}) = \mathrm{Var}(\bar{Y}) + \mathrm{Var}(\hat{\beta}_1 \bar{X}) - 2\mathrm{Cov}(\bar{Y}, \hat{\beta}_1 \bar{X}) \\
&= \frac{\sigma^2}{n} + \frac{\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2}\sigma^2, \text{ since} \\
\mathrm{Cov}(\bar{Y}, \hat{\beta}_1) &= \mathrm{Cov}\left(\frac{1}{n}\sum_{i=1}^n Y_i, \frac{\sum_{i=1}^n (X_i - \bar{X})Y_i}{\sum_{i=1}^n (X_i - \bar{X})^2}\right) \\
&= \frac{1}{n\sum_{i=1}^n (X_i - \bar{X})^2} \sum_{i=1}^n \mathrm{Cov}(Y_i, (X_i - \bar{X})Y_i) \\
&= \frac{1}{n\sum_{i=1}^n (X_i - \bar{X})^2} \sum_{i=1}^n (X_i - \bar{X})\mathrm{Var}(Y_i) = 0.
\end{aligned}
$$

§3.4. (optional) Unbiasedness of $\hat{\sigma}^2$ (1)

First we have

$$
\begin{aligned}
(n-2)\hat{\sigma}^2 &= \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n}(Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \\
&= \sum_{i=1}^{n}[Y_i - \bar{Y} - \hat{\beta}_1(X_i - \bar{X})]^2 \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 + \hat{\beta}_1^2 \sum_{i=1}^{n}(X_i - \bar{X})^2 - 2\hat{\beta}_1 \sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y}) \\
&= \sum_{i=1}^{n}(Y_i - \bar{Y})^2 - \hat{\beta}_1^2 \sum_{i=1}^{n}(X_i - \bar{X})^2. \\
(n-2)E\hat{\sigma}^2 &= E\sum_{i=1}^{n}(Y_i - \bar{Y})^2 - E\hat{\beta}_1^2 \sum_{i=1}^{n}(X_i - \bar{X})^2 \\
&= E\sum_{i=1}^{n}(Y_i - \bar{Y})^2 - [\mathrm{Var}(\hat{\beta}_1) + (E\hat{\beta}_1)^2] \sum_{i=1}^{n}(X_i - \bar{X})^2 \\
&= E\sum_{i=1}^{n}(Y_i - \bar{Y})^2 - \sigma^2 - \beta_1^2 \sum_{i=1}^{n}(X_i - \bar{X})^2.
\end{aligned}
$$

§3.5. (optional) Unbiasedness of $\hat{\sigma}^2$ (2)

Next

$$
\begin{aligned}
& E\sum_{i=1}^{n}(Y_i - \bar{Y})^2 \\
= \quad & E\sum_{i=1}^{n}(\beta_0 + \beta_1 X_i + \epsilon_i - \beta_0 - \beta_1 \bar{X} - \bar{\epsilon})^2 \\
= \quad & E\sum_{i=1}^{n}[\beta_1(X_i - \bar{X}) + (\epsilon_i - \bar{\epsilon})]^2 = \beta_1^2 \sum_{i=1}^{n}(X_i - \bar{X})^2 + E\sum_{i=1}^{n}(\epsilon_i - \bar{\epsilon})^2 \\
& E\sum_{i=1}^{n}(\epsilon_i - \bar{\epsilon})^2 = E\sum_{i=1}^{n}\epsilon_i^2 - 2E\sum_{i=1}^{n}\epsilon_i\bar{\epsilon} + nE\bar{\epsilon}^2 \\
= \quad & n\sigma^2 - nE\bar{\epsilon}^2 = n\sigma^2 - \sigma^2.
\end{aligned}
$$

Finally,

$$(n-2)E\hat{\sigma}^2 = (n-1)\sigma^2 - \sigma^2 = (n-2)\sigma^2,$$

i.e., $E\hat{\sigma}^2 = \sigma^2$.

§3.6. Properties of estimated regression function

- The estimated regression function $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$ is an unbiased estimator of $Y(X) = \beta_0 + \beta_1 X$, i.e., $E\hat{Y} = Y(X)$. It is obvious since $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased.
- The variance of $\hat{Y}$: noting that $\mathrm{Cov}(\bar{Y}, \hat{\beta}_1) = 0$, we have

$$\begin{aligned}
\mathrm{Var}(\hat{Y}) &= \mathrm{Var}(\bar{Y} + \hat{\beta}_1(X - \bar{X})) = \mathrm{Var}(\bar{Y}) + \mathrm{Var}(\hat{\beta}_1(X - \bar{X})) \\
&= \left[\frac{1}{n} + \frac{(X - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]\sigma^2, \\
\text{or} &= \mathrm{Var}(\hat{\beta}_0) + 2X\mathrm{Cov}(\hat{\beta}_0, \hat{\beta}_1) + X^2\mathrm{Var}(\hat{\beta}_1).
\end{aligned}$$

- For the unobserved Y with prediction $\hat{Y}$ at a new value X, the prediction mean square error is given by

$$\begin{aligned}
E(Y - \hat{Y})^2 &= E(\beta_0 + \beta_1 X + \epsilon - \hat{\beta}_0 - \hat{\beta}_1 X)^2 \\
&= \mathrm{Var}(\hat{Y}) + \mathrm{Var}(\epsilon) \\
&= \left[1 + \frac{1}{n} + \frac{(X - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]\sigma^2.
\end{aligned}$$

§3.7. The distribution of the LS estimators

- The normality condition is assumed.
- The $\hat{\beta}_0$ and $\hat{\beta}_1$ are linear transformations of $(Y_1, \ldots, Y_n)$ which is a multivariate normal vector.
- By the property of normal distributions, $\hat{\beta}_0$ and $\hat{\beta}_1$ are normally distributed:

$$\begin{aligned} \hat{\beta}_0 &\sim N\left(\beta_0, \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2}\right]\sigma^2\right), \\ \hat{\beta}_1 &\sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}\right). \end{aligned}$$

- $\hat{\sigma}^2$ is independent from $\hat{\beta}_0$ and $\hat{\beta}_1$. $(n-2)\frac{\hat{\sigma}^2}{\sigma^2}$ follows a χ^2-distribution with degrees of freedom (d.f.) $n-2$. (The verification will be given later when multiple linear regression models are discussed).

§4.1. Significant test

- The significant test is for testing whether or not there is a linear regression relationship between the response variable and the covariate.
- The null hypothesis is that there is no such relation and the alternative hypothesis is the opposite.
- The test statistic is the F-statistic from the ANOVA table:

$$F = \frac{MSR}{MSE}.$$

  Intuitively, it compares the average variation caused by the covariate and that caused by the random error. If those caused by the covariate is much larger, it indicates the existence of the linear relationship.
- Under the null hypothesis, F follows a F-distribution with d.f. 1 and $n-2$ (which will be verified in a more general context later).
- If the observed F value exceeds the upper α quantile $f_{1,n-1}(\alpha)$, the null hypothesis is rejected at level α, otherwise it is not rejected.

§4.2. t-statistic for the inference on β_1

- The inference on β_1 is based on the following t-random variable:

$$T_{\beta_1} = \frac{\hat{\beta}_1 - \beta_1}{s(\hat{\beta}_1)},$$

where $s^2(\hat{\beta}_1) = \hat{\sigma}^2 / \sum_{i=1}^n (X_i - \bar{X})^2$.

- The distribution of T_{β_1} is t_{n-2}, since

$$T_{\beta_1} = \frac{(\hat{\beta}_1 - \beta_1)/\sqrt{\sigma^2 / \sum_{i=1}^n (X_i - \bar{X})^2}}{\sqrt{\hat{\sigma}^2/\sigma^2}},$$

which has the structure $Z/\sqrt{\chi^2_{n-2}/(n-2)}$, with Z and χ^2_{n-1} independent.

§4.3. Two-sided test for β_1

- The hypotheses: $H_0 : \beta_1 = 0$ vs. $H_1 : \beta_1 \neq 0$. The test statistic is given by
$$T_1 = \frac{\hat{\beta}_1}{s(\hat{\beta}_1)}.$$
Under H_0, $T_1 \sim t_{n-2}$.
- The decision rule with significance level α is as follows.

  If $|T_1| > t_{n-2}(\alpha/2)$, reject H_0; otherwise, do not reject H_0, where $t_{n-2}(\alpha/2)$ is the upper $\alpha/2$ quantile of t_{n-2}.
- It is more informative to report the p-value of a test. The p-value for the two-sided t-test is given by $p = P(|t_{n-2}| > |T_1^0|)$, where t_{n-2} denotes the random variable of t-distribution with d.f. $n-2$, T_1^0 is the observed value of T_1.
- Decision rule base on p-value:

  If $p < \alpha$, reject H_0; otherwise, do not reject H_0.
- In the case of simple LRM, the two-sided test on β_1 is equivalent to the significant F-test. In this case, the F statistic equals the square of the t-statistic.

§4.4. One-sided test for β_1

- One-sided test (I)
  - Hypotheses: $H_0 : \beta_1 \leq 0$, vs. $H_1 : \beta_1 > 0$. It is required that the equal sign must be with the null hypothesis.
  - Decision rule: If $T_1 > t_{n-2}(\alpha)$, reject H_0; otherwise, do not reject.
  - p-value: $p = P(t_{n-2} > T_1^0)$.
- One-sided test (II)
  - Hypotheses: $H_0 : \beta_1 \geq 0$ vs. $H_1 : \beta_1 < 0$.
  - Decision rule: If $T_1 < -t_{n-2}(\alpha)$, reject H_0; otherwise, do not reject.
  - p-value: $p = P(t_{n-2} < T_1^0)$.
- Remark: For both two-sided and one-sided tests, the value 0 in the hypotheses can be replaced by any constant c for specific testing problems, the corresponding test statistic is replaced by $(\hat{\beta}_1 - c)/s(\hat{\beta}_1)$.

§4.5. Confidence intervals for β_1

- From the distribution of T_{β_1} we have

$$P\left(-t_{n-2}(\alpha/2) \leq \frac{\hat{\beta}_1 - \beta_1}{s(\hat{\beta}_1)} \leq t_{n-2}(\alpha/2)\right) = 1 - \alpha.$$

- Solving from

$$-t_{n-2}(\alpha/2) \leq \frac{\hat{\beta}_1 - \beta_1}{s(\hat{\beta}_1)} \leq t_{n-2}(\alpha/2),$$

the $100(1-\alpha)\%$ confidence interval for β_1 is obtained as

$$[\hat{\beta}_1 - t_{n-2}(\alpha/2)s(\hat{\beta}_1), \quad \hat{\beta}_1 + t_{n-2}(\alpha/2)s(\hat{\beta}_1)]$$

§4.6. Confidence bounds for β_1

- $100(1-\alpha)\%$ upper confidence bound is given by

$$\beta_1 \leq \hat{\beta}_1 + t_{n-2}(\alpha)s(\hat{\beta}_1),$$

since

$$P(\beta_1 \leq \hat{\beta}_1 + t_{n-2}(\alpha)s(\hat{\beta}_1)) = P\left(\frac{\hat{\beta}_1 - \beta_1}{s(\hat{\beta}_1)} \geq -t_{n-2}(\alpha)\right).$$

- $100(1-\alpha)\%$ lower confidence bound is given by

$$\beta_1 \geq \hat{\beta}_1 - t_{n-2}(\alpha)s(\hat{\beta}_1).$$

since

$$P(\beta_1 \geq \hat{\beta}_1 - t_{n-2}(\alpha)s(\hat{\beta}_1)) = P\left(\frac{\hat{\beta}_1 - \beta_1}{s(\hat{\beta}_1)} \leq t_{n-2}(\alpha)\right).$$

§4.7. Duality between tests and confidence intervals

- Conclusion on two-sided and one-sided hypothesis tests can be drawn from the confidence intervals and confidence bounds respectively. If the hypothesized β_1 value falls outside the $(1-\alpha)\%$ confidence interval or confidence bound, the null hypothesis is rejected at level α, otherwise, it is not rejected.
- The confidence interval corresponds to the two-sided test:
$$H_0 : \beta_1 = c, \ . \ H_1 : \beta_1 \neq c.$$
- The upper confidence bound corresponds to the one-sided test:
$$H_0 : \beta_1 \geq c, vs. \ H_1 : \beta_1 < c.$$
- The lower confidence bound corresponds to the one-sided test:
$$H_0 : \beta_1 \leq c, vs. \ H_1 : \beta_1 > c.$$
- The direction of the bound is the same as the direction of the corresponding alternative hypothesis.

§4.8. The inference on β_0

- The inference on β_0, which is similar to that on β_1, is based on
$$T_0 = \frac{\hat{\beta}_0 - \beta_0}{s(\hat{\beta}_0)}.$$
where
$$s^2(\hat{\beta}_0) = \hat{\sigma}^2 \left[\frac{1}{n} + \frac{\bar{X}^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right].$$
- The distribution of T_0 is also t_{n-2}.

§4.9. Prediction

- Confidence interval of $E(Y)$ when the predictor value is x_h.

$$\hat{y}_h \pm t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{(x_h-\bar{X})^2}{\sum(X_i-\bar{X})^2}\right)}$$

  where $\hat{y}_h$ is the "predicted value" of the response when the predictor is x_h.

- Prediction interval of Y_{new} when the predictor value is x_h.

$$\hat{y}_h \pm t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1+\frac{1}{n}+\frac{(x_h-\bar{X})^2}{\sum(X_i-\bar{X})^2}\right)}$$

- R code

```
predict(object, newdata, interval = c("none",
"confidence", "prediction"), level = 0.95)
```

§4.10. MM example: hypotheses

- The study is to investigate whether or not a person's muscle mass is to decrease with age, which is equivalent to whether or not $\beta_1 < 0$. That $\beta_1 < 0$ is to be discovered (or confirmed by the study).
- By the guideline for formulating the hypotheses, the hypotheses are formulated as:
$$H_0 : \beta_1 \geq 0, \text{ vs. } H_1 : \beta_1 < 0.$$
- The summary fitted results are as follows,

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 148.0507    11.5629  12.804 4.05e-09 ***
age          -1.0236     0.1882  -5.439 8.72e-05 ***
Residual standard error: 8.344 on 14 degrees of freedom
Multiple R-squared:  0.6788,    Adjusted R-squared:  0.6559
F-statistic: 29.59 on 1 and 14 DF,  p-value: 8.721e-05
```

§4.11. MM example: hypothesis testing

- From the summary results, $T_1 = -5.439$, and the d.f. of the test statistic is 14. Let the level of the test $\alpha = 0.05$.
- The upper 0.05-quantile $t_{14}(0.05)$ is computed either by `qt(0.95,14)` or `qt(0.05,14,lower.tail=FALSE)`, which gives 1.76131. Since $T_1 = -5.439 < -1.76131$, H_0 is rejected and it is confirmed that the muscle mass decreases with age.
- The p-value is computed by `pt(-5.439,14)`, which gives $4.36e{-}05$. H_0 is rejected since the p-value is smaller than 0.05.
- Note that for the one-sided test for either $H_0 : \beta_1 \geq 0$ or $H_0 : \beta_1 \leq 0$, the p-value is half of that for the two-sided test which is given in the summary results.
- A 95% upper confidence bound (corresponding to the direction of the alternative hypothesis) for β_1 is constructed as

$$\beta_1 \leq -1.0236 + 0.1882 * 1.76131 = -0.6921.$$

  The hypothesized value $\beta_1 = 0$ is outside the bound, so based on the upper bound, the hypothesis $H_0 : \beta_1 \geq 0$ is rejected at level 0.05.

§4.12. MM example: prediction

- Do prediction for a person with age 65:

```
> predict(MM.fit,list(age=65),interval="confidence",level=0.95)
       fit      lwr      upr
 81.51737 76.67932 86.35542
>  predict(MM.fit,list(age=65),interval="prediction",level=0.95)
       fit      lwr      upr
 81.51737 62.97935 100.0554
```

- In the function `predict`, the argument `newdata` must be a list object, the variable name in the list must be the same as those in the formula specification within `lm`. Recall that, for the MM example, the formula specification is as follows:

```
MM.fit=lm(mass~age)
```

§4.13. Birth Rate (BR) Example

This example concerns the Teen Birth Rate and Poverty Level of the 50 states and the District of Columbia in the United States. The dataset consist of 51 observations. The variables are `TeenBrth` = year 2002 birth rate per 1000 females 15 to 17 years old and `PovPct` = poverty rate, which is the percent of the state's population living in households with incomes below the federally defined poverty level. (Data source: Mind On Statistics, 3rd edition, Utts and Heckard). The data file `poverty.txt` is provided in Canvas.
The investigator wants to find out whether there is any relation between the teen birth rate and the poverty rate in the consideration of social policies.
If we were the investigator, we should proceed as follows:

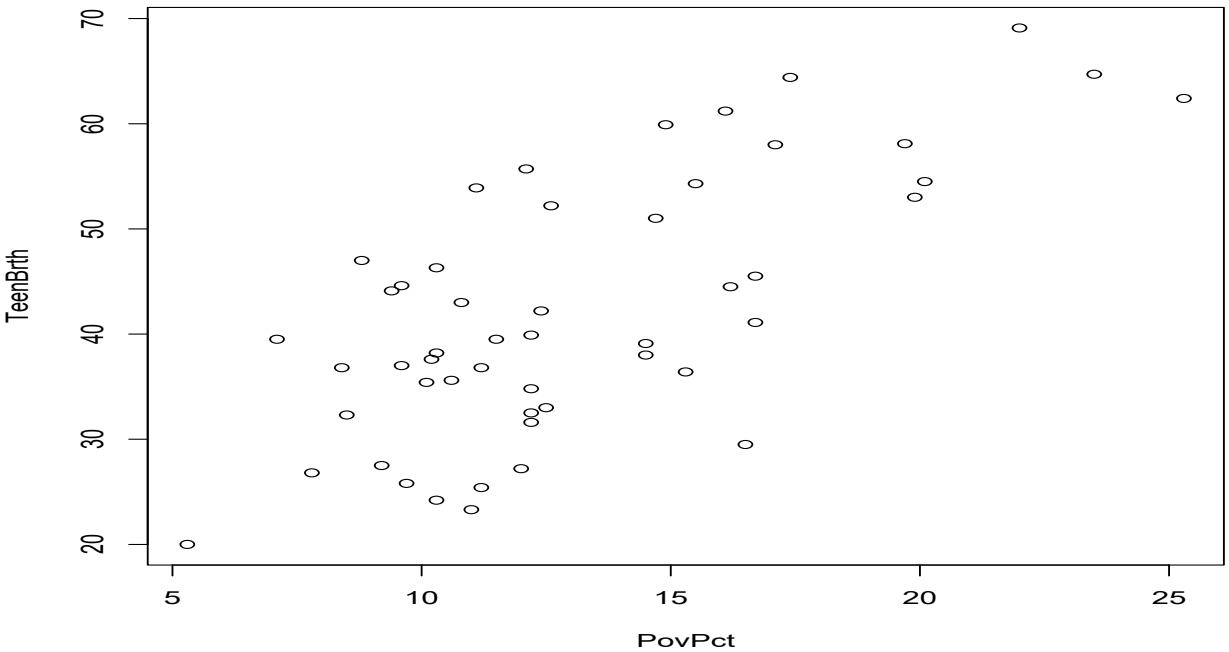
Step I . Establish a regression model: (1) take teen birth rate as the response variable and the poverty rate as the covariate; (2) plot the response against the covariate to get a rough idea on what the relation might be; (3) fit a regression model, (4) check the adequacy of the model.

Step II . Formulate the problems into statistical questions and do the analysis.

§4.14. R code for Step I

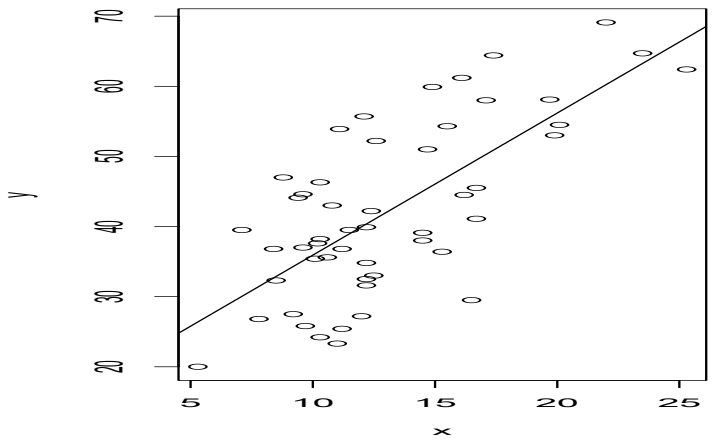
```
# Input data and make variable available
   BR.dat=read.table(file="D://Rsession/poverty.txt")
   y=BR.dat$TeenBrth; x=BR.dat$PovPct
# Draw scatter plot
   plot(x, y)
# Fit regression model
   BR.fit=lm(y~x); summary(BR.fit)
# Model adequacy checking
   yhat=BR.fit$fitted; r=BR.fit$resid;
   par(mfrow=c(1,2))
   plot(x, y)
   abline(BR.fit)
   plot(yhat, r,"Residual plot")
```

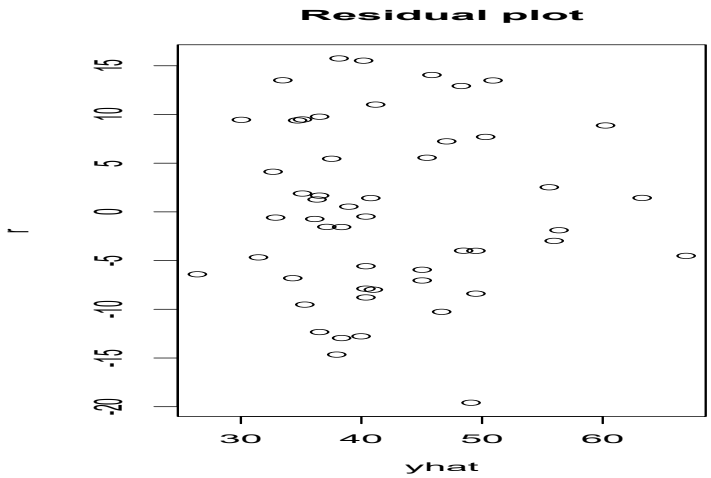
§4.15. Scatter plot of TeenBrth against PovPct



The plot shows a upward linear trend, which suggests a linear relation between teen birth rate and poverty rate.

§4.16. Plots for checking model adequacy





The left is the scatter plot with the fitted regression line $\hat{y} = 15.6737 + 2.0255x$ imposed. Both plots do not show any inadequacy of the simple LRM.

§4.17. Statistical analysis

```
Fitted results:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  15.6737     4.0318   3.888 0.000305 ***
x             2.0255     0.2925   6.925 8.67e-09 ***
Residual standard error: 8.846 on 49 degrees of freedom
Multiple R-squared:  0.4946,    Adjusted R-squared:  0.4843
F-statistic: 47.95 on 1 and 49 DF,  p-value: 8.67e-09

Varaince-covarince matrix:
            (Intercept)       x
(Intercept)     16.2553 -1.1222
x               -1.1222  0.0856
```

Given the above results,

(i) conduct a significant test of the regression model.

(ii) obtain the proportion of the variation in teen birth rate which is explained by poverty rate.

(iii) Test the hypotheses: $H_0 : \beta_1 \leq 0$ vs. $H_1 : \beta_1 > 0$.

(iv) construct a 95% confidence lower bound for β_1.

(v) construct a 95% prediction interval for the teenage birth rate for a state with poverty rate 30.

§4.18. Solution

(i) Conduct a significant test of the regression model.

The test statistic for the significant test is $F = MSR/MSE$. *Under the hypothesis of non-significance,* $F \sim F_{1,49}$. *The value of* F *is obtained from the fitted results as 47.95, its p-value as* `8.67e-09`. *The p-value is smaller than any reasonable level, it confirms that the regression relation is significant.*

(ii) obtain the proportion of the variation in teen birsth rate which is explained by poverty rate.

The proportion is given by R^2, *which is obtained from the fitted results as 0.4946.*

(iii) Test the hypotheses: $H_0 : \beta_1 \leq 0$ vs. $H_1 : \beta_1 > 0$.

The test statistic from the summary results is $T_1 = 6.925$. *The p-value is* $P(t_{49} > 6.925) =$ `4.334208e-09`, *noting that it is half of the p value* `8.67e-09` *for two-sided test. The null hypothesis is rejected.*

§4.19. Solution (cont.)

(iv) construct a 95% confidence lower bound for β_1.
The confidence lower bound is constructed as

$$\beta_1 \geq \hat{\beta}_1 - sd(\hat{\beta}_1)t_{49}(0.05) = 2.0255 - 0.2925 * 1.67655 = 1.5351.$$

(v) construct a 95% prediction interval for the teenage birth rate for a state with poverty rate 30.
The prediction interval is given by the formula

$$\hat{y}_h \pm t_{49}(0.025)\sqrt{Var(Y_{new})}$$

where $Var(Y_{new}) = Var(\hat{y}_h) + \hat{\sigma}^2$, $\hat{y}_h = \hat{\beta}_0 + \hat{\beta}_1 x_h$, *and* $Var(\hat{y}_h)$ *can be computed by*

$$\begin{aligned} Var(\hat{y}_h) &= Var(\hat{\beta}_0) + x_h^2 Var(\hat{\beta}_1) + 2x_h Cov(\hat{\beta}_0, \hat{\beta}_1) \\ &= 16.2554 + 30^2 * 0.0856 + 2 * 30 * (-1.1222) = 25.9634. \end{aligned}$$

Hence, $Var(Y_{new}) = 25.9634 + 8.846^2 = 104.2151.$

§4.20. Solution (cont.)

(v) *(cont.)* $\hat{y}_h = 15.6737 + 2.0255 * 30 = 76.4387$, $t_{49}(0.025) = 2.0096$. $\sqrt{Var(Y_{new})} = \sqrt{104.2151} = 10.20858$. *The prediction interval is computed as*

$$\begin{aligned} & [76.4387 - 10.20858 * 2.0096,\ 76.4387 + 10.20858 * 2.0096] \\ = \quad & [55.92354,\ 96.95386] \end{aligned}$$

The prediction interval can be computed by the R code `predict(BR.fit,list(x=30),interval="prediction", level = 0.95)` *which yields the interval*

$$[55.92711,\ 96.94874].$$

The difference is caused by rounding error.

§4.21. Conclusion on the BR example

- There is a significant linear relation between the teen birth rate and poverty rate.
- The teen birth rate increases as poverty rate increases. With 95% confidence, when poverty rate increases by a unit, the teen birth rate increases by at least 1.5351.

A brief summary

- The simple LRM is introduced and its properties and analysis are discussed in this lecture notes.
- The main concepts and problems such as least square estimation, coefficient of determination, significant test, two-sided and one-sided tests, confidence intervals and bounds, prediction, etc. are discussed in details.
- The concepts and problems of simple LRM can be readily extended to multiple linear regression models which will be discussed in Lecture notes 2.